WIGVO: Real-Time Bidirectional Speech Translation over Legacy PSTN Calls via Dual-Session Echo Gating

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Abstract

Real-time speech translation with large language models (LLMs) has become feasible in controlled settings such as mobile apps and web browsers, where developers assume wideband audio and client-side echo cancellation. Recent end-to-end full-duplex models further push latency below 200 ms but remain confined to broadband environments. However, deploying such systems over the Public Switched Telephone Network (PSTN) remains challenging due to narrowband G.711 audio, unpredictable round-trip delays, and absence of client-side signal processing. We present **WIGVO**, a server-side relay system that enables bidirectional LLM-based speech translation over ordinary telephone calls without requiring app installation or carrier integration. A central contribution is addressing what we term *echo-induced self-reinforcing translation loops*: synthesized speech echoing back through the PSTN gets re-ingested and repeatedly translated. WIGVO solves this through a dual-session architecture with deterministic silence injection and energy-based voice activity detection (VAD) gating. We evaluate WIGVO on 96 completed PSTN calls (37 fully instrumented) across three communication modes—voice-to-voice (V2V), text-to-voice (T2V), and full-agent—observing 562 ms median caller-to-callee latency and 1,994 ms median callee-to-caller latency, zero echo-induced translation loops, and USD 0.42 per minute cost. The system is deployed at https://wigvo-web-gzjzn35jyq-du.a.run.app with a live demo and video walkthrough.

1 Introduction

Streaming speech-to-speech translation has advanced rapidly with Realtime LLM APIs (OpenAI, 2024), but existing work focuses on WebRTC and VoIP settings where wideband audio and client-side acoustic echo cancellation (AEC) (ITU-T, 2015) are available (Seamless Communication et al., 2023a; Radford et al., 2023). The Public Switched Telephone Network (PSTN), however, remains the primary inbound telephony interface for local businesses, hospitals, and government offices (International Telecommunication Union, 2020). PSTN encodes audio via G.711 μ-law at 8 kHz (ITU-T, 1988), introduces end-to-end delays of 80–600 ms (ITU-T, 2003), and provides no client-side AEC. For foreign residents, people with speech anxiety, or users with hearing impairments, making such calls remains a significant barrier.

Recent end-to-end full-duplex speech models such as Moshi (Défossez et al., 2024) and Hibiki (Labiausse et al., 2025) achieve sub-200 ms latency in broadband settings, but they do not address PSTN-specific signal processing challenges: narrowband codec artifacts, telephony echo without client-side AEC, and VAD failures on noisy G.711 audio.

We present **WIGVO**, a server-side relay that bridges a web client and a standard PSTN phone number through two concurrent LLM-backed streaming sessions. The caller speaks (or types) via a browser; the callee answers on an ordinary phone—no app installation, carrier integration, or special hardware is required on either end. WIGVO translates bidirectionally in real time, supporting voice-to-voice (V2V), text-to-voice (T2V), and full-agent modes.

A key challenge is what we call an **echo-induced self-reinforcing translation loop**: TTS audio played to the callee echoes back through the PSTN, is re-recognized as new speech, and gets translated again—creating a runaway feedback cycle. This problem is specific to full-duplex telephony channels and cannot be solved by VAD threshold tuning alone. Our solution operates at the architectural level: (i) strict directional separation via two independent Realtime sessions, (ii) deterministic silence injection during TTS playback

windows, and (iii) energy-based gating calibrated for μ-law dynamics.

We evaluate WIGVO on 96 completed PSTN calls and report latency, echo suppression, and cost metrics. Our contributions are:

- We formalize the *echo-induced translation loop* problem in streaming speech translation over telephony.
- We propose a *dual-session gated architecture* combining directional separation, silence injection, and energy-based gating for PSTN environments.
- We deploy and evaluate a working relay server across three communication modes, achieving zero echo-induced loops across 96 real PSTN calls with 562 ms median caller-to-callee latency at USD 0.42 per minute.

2 Related Work

Simultaneous speech translation. Recent systems such as SeamlessM4T (Seamless Communication et al., 2023b) and Seamless Streaming (Seamless Communication et al., 2023a) achieve real-time speech translation with expressive and multilingual models. ESPnet-ST-v2 (Yan et al., 2023) provides a comprehensive toolkit for offline and simultaneous translation. However, all assume wideband audio inputs from controlled environments and do not address PSTN-specific challenges such as G.711 codec artifacts, telephony echo, or narrowband VAD.

End-to-end full-duplex speech models. Recent work has moved beyond cascaded pipelines toward end-to-end full-duplex architectures that eliminate explicit turn boundaries. Moshi (Défossez et al., 2024) is the first real-time full-duplex spoken LLM, modeling user and system speech as parallel token streams with a practical latency of 200 ms. PersonaPlex (Roy et al., 2026) fine-tunes this architecture with hybrid voice and role conditioning, reducing smooth turn-taking latency to 70 ms and surpassing commercial systems including Gemini Live on conversational naturalness. For speech-to-speech translation specifically, Hibiki (Labiausse et al., 2025) applies the same multistream architecture to simultaneous interpretation, achieving near-human-interpreter quality on French-to-English translation in broadband settings. Notably, Hibiki's limitation to a single language pair underscores the difficulty of scaling end-to-end S2ST systems—a challenge that WIGVO sidesteps through its modular cascade design, which enables language extensibility by swapping individual components. However, all these systems assume clean, wideband (24 kHz) audio. WIGVO addresses a complementary and underexplored setting: real-time S2ST over PSTN, where G.711 μ-law encoding at 8 kHz, acoustic echo, and codec-induced VAD failures require architectural intervention that end-to-end models do not provide.

Telephony AI agents. Google Duplex (Leviathan and Matias, 2018) demonstrated autonomous PSTN calls for restaurant reservations, but performs task completion in a single language rather than bidirectional translation. Commercial platforms such as Vapi[1] and Bland.ai[2] provide LLM-powered phone agents but focus on monolingual voice assistants without cross-lingual translation capabilities.

Relay services for accessibility. Telecommunications Relay Services (TRS), mandated by the FCC in the United States, provide human operators who relay calls for deaf and hard-of-hearing users (Federal Communications Commission, 2024). These services are limited to specific countries, require human intermediaries, and do not support cross-lingual translation.

Accessibility-oriented translation. sign.mt (Moryossef, 2024) provides bidirectional sign language translation with an accessibility focus; our work targets telephony and addresses echo-induced loops. Table 1 positions WIGVO against these systems.

3 System Architecture

Figure 1 shows the high-level architecture. A browser-based client connects to the relay server via WebSocket, sending 16 kHz PCM audio and receiving translated audio, captions, and status events. The relay manages two independent Realtime LLM sessions and a telephony gateway connection.

AudioRouter and pipelines. An `AudioRouter` implements the Strategy pattern to dispatch events to one of three pipeline implementations: V2V,

[1] https://vapi.ai
[2] https://bland.ai

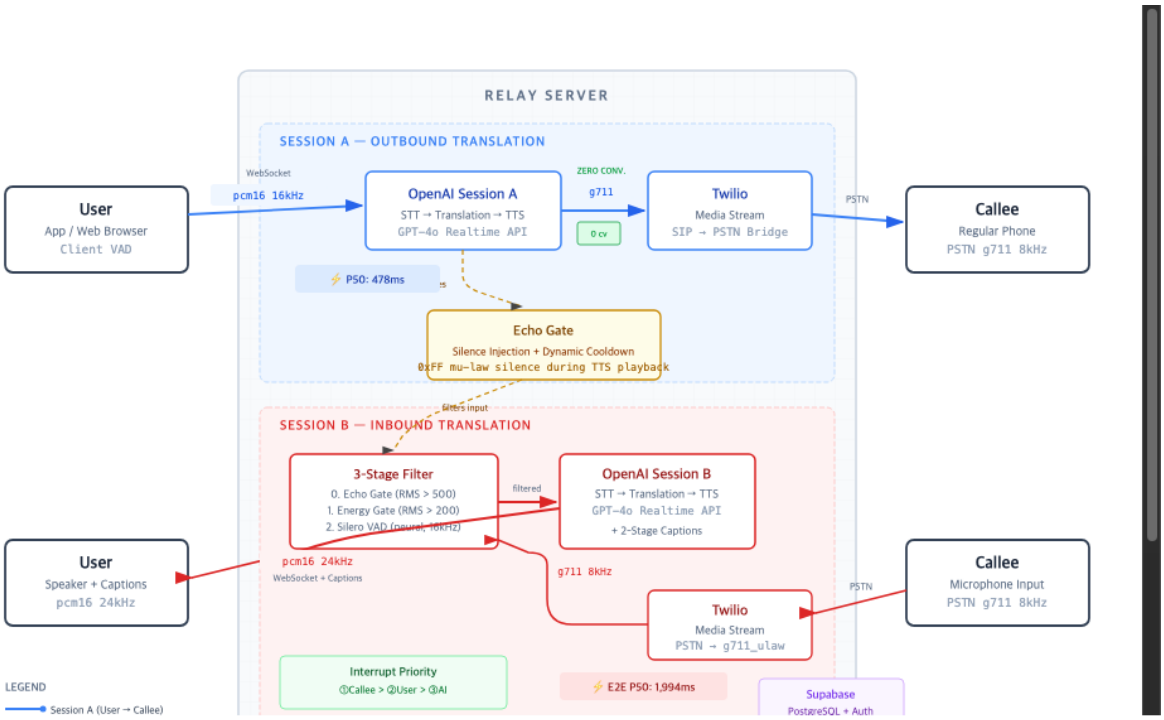

Figure 1: WIGVO system architecture. Session A (blue) translates user speech and outputs G.711 directly to Twilio; Session B (red) receives PSTN audio through a 3-stage filter pipeline (Echo Gate → Energy Gate → Silero VAD). The Echo Gate (dashed) prevents feedback loops by injecting μ-law silence during TTS playback.

System	PSTN	Bidir.	S2S	Echo	A11y
SeamlessM4T	×	✓	✓	N/A	×
Moshi / Hibiki	×	×	✓	N/A	×
Google Duplex	✓	×	×	N/D	×
FCC TRS	✓	✓	×	Human	✓
sign.mt	×	✓	✓	N/A	✓
Vapi / Bland	✓	×	×	N/D	×
WIGVO	✓	✓	✓	✓	✓

Table 1: Comparison with existing systems. PSTN: works over telephone network; Bidir.: bidirectional translation; S2S: automated speech-to-speech (not human-mediated); Echo: handles telephony echo; A11y: accessibility modes. N/D: not disclosed.

T2V, or Full-Agent. Each pipeline manages the lifecycle of two Realtime sessions and cross-cutting components (guardrails, context window, metrics).

Dual Realtime sessions. Session A receives browser audio (PCM16) and produces translated G.711 μ-law for the telephony gateway. Session B receives PSTN audio (G.711 μ-law, 8 kHz) and produces translated audio and captions for the browser. Each session has its own system prompt, sliding context window (6 turns, empirically chosen to balance translation coherence and token cost), and event stream, ensuring strict directional separation.

Communication modes. **V2V**: both directions use streaming ASR+TTS. **T2V**: the caller types text, which is converted to TTS for the callee; the callee's speech is transcribed and displayed as captions. **Full-Agent**: the LLM autonomously conducts the call based on collected user intent.

4 Echo Gating Mechanisms

4.1 The Echo Loop Problem

In a naive single-session design, both speakers' audio feeds into one streaming session. TTS output played to the callee echoes back through the PSTN after 80–600 ms, gets re-recognized, and triggers another translation cycle. We observed this reliably in early prototypes: the system would generate progressively distorted paraphrases of its own output until manually interrupted.

We initially attempted a Pearson-correlation echo detector comparing outgoing TTS buffers

with incoming audio. While effective in controlled tests, it failed in production due to μ-law nonlinear quantization, variable delays, and background noise producing spurious correlations. This detector is now disabled.

A single-session baseline without gating produced echo loops in 8 of 10 test calls; the Pearson-correlation detector reduced loops to 3 of 10 but introduced frequent false positives that blocked legitimate callee speech. Echo Gate v2 eliminated loops entirely (0 of 96) while preserving callee interruption capability.

4.2 Echo Gate v2: Silence Injection

Our production solution, **Echo Gate v2**, operates deterministically:

1. When Session A begins streaming TTS to the PSTN, an *echo window* is activated. Duration is estimated from byte length and sample rate, empirically capped at 1.2 s to cover typical single-utterance TTS outputs.

2. During this window, incoming PSTN audio is replaced with μ-law silence (0xFF) before forwarding to Session B. This blocks echo at the server while allowing the Realtime API's VAD to naturally detect end-of-speech.

3. A dynamic cooldown of 0.3 s (0.5 s for T2V mode) accounts for PSTN jitter before resuming normal forwarding. An additional 2.0 s post-echo settling window suppresses AGC recovery noise while permitting high-energy breakthrough ($\geq$400 RMS) for genuine callee speech.

While comfort noise generation (RFC 3389; Zopf 2002) addresses perceptual continuity for human listeners in VoIP, we identify a distinct failure mode: echo-suppression-induced audio gaps cause streaming VAD state machine deadlock in neural speech APIs. Specifically, if the relay simply *drops* audio during echo windows rather than injecting silence, the Realtime API's server-side VAD never observes a silence-to-speech transition and remains stuck in a "speaking" state indefinitely. To our knowledge, this failure mode and its mitigation via silence injection have not been reported in the context of neural streaming S2ST pipelines.

Across 96 completed calls, Echo Gate v2 produced zero echo-induced translation loops. Echo gate activated 127 times total (3.4/call on average), with 0 breakthrough incidents, confirming that silence injection effectively prevents self-reinforcing translation cycles.

4.3 Energy-Based Gating

RMS energy monitoring provides a secondary gate. During echo windows, only high-energy signals ($\geq$400 RMS) break through as genuine callee speech; typical echo energy (100–400 RMS) remains suppressed. Outside echo windows, a lower threshold (150 RMS) filters line noise without clipping normal speech. This two-tier scheme allows callee interruptions to override the echo gate immediately. A three-level interrupt priority ensures callee speech immediately cancels active TTS: the relay prioritizes callee audio over caller input, and caller input over ongoing AI-generated TTS, enabling natural turn-taking even during active translation.

5 PSTN-Aware VAD and Robustness

The echo gate described in Section 4 suppresses self-translation loops but does not address a second class of failures: the Realtime API's server-side VAD itself malfunctions on narrowband PSTN audio. This section describes these failures and the two-stage local VAD that mitigates them. Together, the echo gate (Section 4) and local VAD form a unified inbound audio pipeline: PSTN audio first passes through echo gating and energy-based filtering, then through neural VAD, before reaching Session B. Figure 2 illustrates this pipeline and its temporal behavior.

5.1 Server VAD Failure Modes on PSTN

OpenAI's Realtime API (OpenAI, 2024) provides a server-side VAD that works well for wideband WebRTC audio. On PSTN lines, however, we observed three failure modes:

1. **Background noise misclassification.** Constant low-level PSTN noise (line hum, codec quantization artifacts) is classified as speech, causing the VAD to enter a "speaking" state that never terminates. We observed stuck durations ranging from 15 s to 72 s before manual intervention.

2. **Delayed `speech_stopped` events.** Even after genuine speech ends, the noisy PSTN floor prevents the energy level from dropping

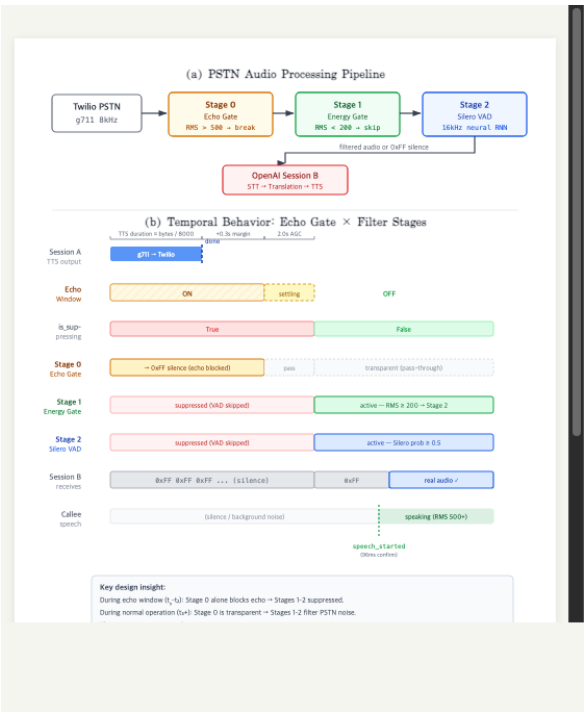

Figure 2: PSTN audio processing pipeline. (a) Three-stage filter between Twilio input and Session B. (b) Temporal behavior: during TTS playback, the Echo Gate replaces audio with μ-law silence while Stages 1–2 are suppressed; after settling, Stage 0 becomes transparent and Stages 1–2 filter PSTN noise.

below the server VAD's silence threshold, delaying end-of-utterance detection by 15–72 s.

3. **Audio discontinuity deadlock.** When the relay drops audio frames (e.g., during echo suppression), the server VAD never observes a clean silence interval and fails to emit a `speech_stopped` event entirely.

5.2 Two-Stage Local VAD

To address these failures, WIGVO implements a two-stage local VAD on the relay server, applied to incoming PSTN audio before it reaches the Realtime API:

Stage 1: RMS Energy Gate. A fast energy check filters sub-speech signals. We set the threshold at 200 RMS based on empirical μ-law noise distributions: background noise typically falls in the 50–200 RMS range, while speech aligns around 500–2,000+ RMS. Frames below 200 RMS are classified as noise and replaced with silence, preventing them from reaching Stage 2.

Stage 2: Silero VAD. Frames passing the energy gate are processed by Silero VAD (Silero Team, 2021), a lightweight neural voice activity detector. Since Silero expects 16 kHz input but PSTN audio arrives at 8 kHz, we apply zero-order hold (ZOH) upsampling—chosen over linear interpolation for its lower computational cost and deterministic latency (no lookahead required)—before inference. Each frame is 32 ms (512 samples at 16 kHz).

Hysteresis state machine. To avoid rapid state oscillation, we apply asymmetric hysteresis thresholds:

- **SILENCE→SPEAKING**: probability ≥ 0.5 for 3 consecutive frames (96 ms). Fast onset preserves responsiveness.
- **SPEAKING→SILENCE**: probability < 0.35 for 15 consecutive frames (480 ms). Slow offset avoids premature cutoff during natural pauses.

This design reduces `speech_stopped` latency from 15–72 s (server VAD on PSTN) to a consistent 480 ms, a reduction of over 96%.

5.3 STT Hallucination Blocklist

Whisper (Radford et al., 2023) occasionally generates broadcast-style hallucinations on low-energy PSTN input—for example, Korean news sign-off phrases such as "MBC News" or "Thank you for watching." A 15-pattern blocklist intercepts these outputs before they reach the translation stage; 5 hallucinated outputs were blocked across 37 instrumented calls.

6 Evaluation

We evaluated WIGVO on 96 PSTN calls (37 fully instrumented) for Korean↔English translation. The remaining 59 calls completed successfully but lacked full per-turn instrumentation due to incremental deployment of the metrics collection pipeline.

6.1 Latency

Table 2 reports latency statistics across the 37 instrumented calls.

Session A achieves 562 ms median latency, within the range for interactive communication. Session B, which depends on PSTN audio quality and streaming ASR, shows 1,994 ms median. STT (Whisper) accounts for 74.7% of Session B mean latency across 103 paired turns, with translation contributing the remainder. Figure 3 shows the

Stage	Avg	P50	P95	Max	N
Session A (ms)	617	562	1,023	2,256	141
Session B E2E (ms)	2,249	1,994	5,390	10,503	207
STT (ms)	2,243	1,969	5,116	—	189
Translation (ms)	763	457	2,255	—	103
First message (ms)	2,585	1,550	7,460	11,308	31

Table 2: Translation latency across 37 instrumented calls. Session A: caller→callee (ASR+translate+TTS). Session B: callee→caller, decomposed into STT (Whisper) and translation (GPT-4o) components. N varies across rows because each metric has a different measurement scope: direction-specific turn counts, successful STT completions, and paired STT+E2E availability for translation decomposition.

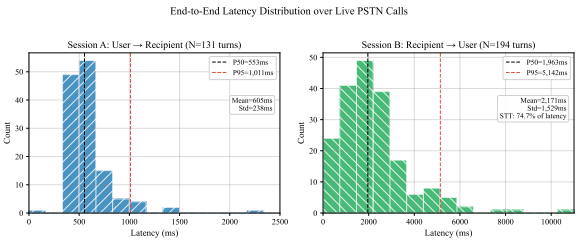

Figure 3: End-to-end latency distributions for Session A (caller→callee, N=141 turns) and Session B (callee→caller, N=207 turns) over live PSTN calls.

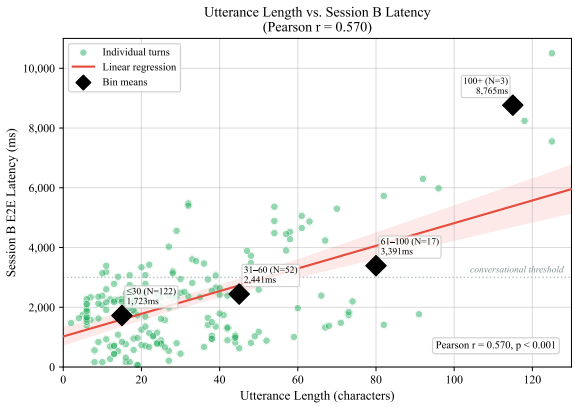

Figure 4: Utterance length (characters) vs. Session B E2E latency. Pearson $r = 0.529$ ($p < 0.001$), indicating that longer recipient utterances incur higher ASR-dominated latency.

end-to-end latency distributions for Session A and Session B. Figure 4 shows the correlation between utterance length and Session B latency ($r = 0.529$, $p < 0.001$). The five highest Session B outliers (5,981–10,503 ms) all correspond to long callee utterances of 92–125 characters, where streaming STT accounts for ≥95% of E2E latency; the translation step in these cases remains under 1,500 ms. This confirms that tail latency is driven by utterance length and ASR processing time, not architectural overhead.

T2V mode shows lower Session A latency (no caller-side ASR), while Agent mode shows higher Session B latency due to function-calling overhead.

While Session B's ∼2 s median exceeds ITU-T G.114's 150 ms one-way threshold for conversational speech (ITU-T, 2003), that standard applies to same-language dialogue, not cross-lingual translation where additional processing is inherent. A more appropriate baseline is professional simultaneous interpretation, where ear-voice span—the delay between hearing source speech and producing the target—ranges from 2 to 5 s (Goldman-Eisler, 1972; Timarová et al., 2011). WIGVO's Session B latency falls at the lower bound of this range. Moreover, the latency is structurally asymmetric by design: Session A (562 ms) keeps the caller-to-callee direction within interactive thresholds, while Session B prioritizes STT accuracy on degraded G.711 audio over speed. Decomposing Session B, STT accounts for 74.7% of mean latency; the translation step itself is 457 ms at P50. This indicates that the bottleneck is narrowband ASR, not the relay architecture.

6.2 Echo and Safety

Table 3 reports echo suppression, VAD, and guardrail statistics across the 37 instrumented calls.

The echo gate activated 3.4 times per call on average, corresponding to TTS playback events. Critically, **zero echo-induced translation loops** were observed, whereas early prototypes without gating reliably produced such loops. VAD false triggers (1.6/call) were mitigated by a minimum speech duration filter (480 ms hysteresis). The guardrail module blocked 5 hallucinated outputs (broadcast-style phrases from Whisper on low-energy input).

6.3 Cost

Across the 37 instrumented calls (45.5 minutes total), WIGVO consumed 638,035 tokens at a total cost of USD 19.05, yielding **USD 0.42/min** (USD 0.50/call). These costs reflect dual-session Realtime API pricing and remain an order of magnitude cheaper than professional telephone inter-

Metric	Total	Per-call avg
Echo gate activations	127	3.4
Echo-induced loops	0	0.0
VAD false triggers	59	1.6
Hallucinations blocked	5	0.13

Table 3: Echo, VAD, and guardrail statistics over 37 instrumented calls. Zero echo loops were observed.

System	Network	Bidir.	Latency
Seamless Str.	Wideband	×	<2 s AL
Moshi	Wideband	×†	∼200 ms
Hibiki	Wideband	×	2.8–5.6 s LAAL
Google Duplex	PSTN	×‡	N/D
Vapi	WebRTC/PSTN	×‡	∼965 ms*
Bland.ai	PSTN	×‡	<400 ms*
WIGVO SA	**PSTN**	✓	**562 ms P50**
WIGVO SB	**PSTN**	✓	**1,994 ms P50**

Table 4: Latency comparison with related systems. AL: Average Lagging; LAAL: Length-Adaptive AL. †: full-duplex but monolingual (no translation). ‡: monolingual voice agent (no cross-lingual). ∗: vendor-reported; independent reviews report higher values. N/D: not disclosed. Bidir.: bidirectional cross-lingual translation.

pretation services (USD 1–3/min).[3]

6.4 Baseline Comparison

Table 4 contextualizes WIGVO's latency against related systems. Direct comparison is inherently limited because no existing system performs bidirectional cross-lingual speech translation over PSTN: research systems operate in wideband settings with incompatible latency metrics (Average Lagging, frame-level), while commercial telephony platforms are monolingual voice agents.

Research S2ST systems report latency on wideband (16–24 kHz) benchmarks. Hibiki (Labiausse et al., 2025) achieves 2.8 s LAAL on short-form French→English but is limited to a single language pair; Seamless Streaming (Seamless Communication et al., 2023a) covers 36 target languages at <2 s AL but does not address telephony. Moshi (Défossez et al., 2024) achieves 200 ms full-duplex latency but performs monolingual dialogue, not translation. None of these systems handle G.711 μ-law, PSTN echo, or narrowband VAD.

Commercial telephony platforms (Vapi, Bland.ai) operate over PSTN but provide monolingual voice agents without translation. Twilio's industry benchmark for AI voice agents targets 1,115 ms median mouth-to-ear turn gap (Twilio, 2025). WIGVO's Session A (562 ms) falls well within this target. Session B (1,994 ms) exceeds it, but this direction includes an additional cross-lingual translation step that monolingual agents do not perform.

[3]Based on published per-minute rates for over-the-phone interpretation (OPI) from major commercial providers.

Figure 5: WIGVO web interface during a V2V call. Left: chat and caption view with bidirectional translations. Right: call status, duration, and mode indicator.

7 Demonstration

WIGVO is deployed at https://wigvo-web-gzjzn35jyq-du.a.run.app with a live demo accessible to reviewers.[4] The source code is publicly available at https://github.com/wigtn/wigvo-v2. The web interface supports scenario selection (restaurant, hospital, government office), real-time caption display, and mode switching. Figure 5 shows the interface during an active V2V call.

We demonstrate two scenarios:

Restaurant reservation (V2V). An English-speaking user calls a Korean restaurant. Session A translates caller utterances into Korean TTS for the callee; Session B returns Korean responses as English audio and captions.

Medical appointment (T2V). A Korean user with speech anxiety types messages that WIGVO speaks to a clinic via TTS; the clinic's spoken responses appear as real-time captions. A typing indicator is relayed to the callee during text composition.

8 Discussion

Scope of contribution. WIGVO's contribution is architectural: demonstrating that existing Realtime LLM APIs can be made robust for PSTN

[4]Demo video: https://youtu.be/PLACEHOLDER

environments through dual-session separation, deterministic echo gating, and energy-based filtering. Translation quality inherits the capabilities of GPT-4o Realtime (OpenAI, 2024); formal evaluation (BLEU, chrF) against independent reference translations is planned for the camera-ready version.

Comparison limitations. As shown in Table 4, no existing system occupies the same design point as WIGVO—bidirectional cross-lingual S2ST over narrowband PSTN. Research systems target wideband settings with different metrics, and commercial telephony platforms are monolingual. We therefore report absolute latency and echo metrics rather than head-to-head comparisons, which would require re-implementing competitors in a PSTN environment or restricting WIGVO to wideband—neither of which would yield a fair comparison.

Broader applications. The dual-session gated architecture is applicable beyond translation: multilingual call centers, voice inclusion services for users with disabilities, and telephony-native LLM agents that combine task completion with cross-lingual communication.

9 Limitations and Future Work

Language and environment coverage. The current evaluation focuses on Korean-English over PSTN. WIGVO's modular cascade architecture is inherently language-agnostic, and multilingual extension requires no architectural changes beyond model substitution. In broadband VoIP settings, the echo gate and ZOH upsampling become unnecessary, and the system degrades gracefully to a standard cascade pipeline. Systematic multilingual evaluation and mixed PSTN-VoIP benchmarking remain future work.

Session B latency reduction. Replacing Whisper STT (74.7% of Session B latency) with a self-hosted streaming model (Kyutai, 2025) or an end-to-end full-duplex model (Défossez et al., 2024; Labiausse et al., 2025) could reduce Session B from ∼2 s to under 800 ms, bringing it below the Twilio industry benchmark of 1,115 ms (Twilio, 2025). The PSTN relay layer, echo gate, and VAD architecture would remain necessary regardless, as they address network-level constraints orthogonal to model design.

Cost and user evaluation. Per-minute cost (USD 0.42) is driven by Realtime API audio token pricing; self-hosted STT alternatives could reduce this. A formal user study (e.g., SUS assessment; Brooke 1996) with target populations has not yet been conducted and is planned as future work.

Ethics Statement

All calls were initiated by the authors and collaborators for system testing. Audio was logged for debugging and aggregate metrics only, with no personally identifiable information retained beyond temporary phone numbers. For deployment with real users, informed consent, data minimization, and compliance with privacy regulations (e.g., GDPR, PIPA) are required. The accessibility use case (T2V for speech anxiety) aims to reduce communication barriers but should be deployed with user agency and opt-in participation.

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